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标题	Application of the digital annealer unit in optimizing chemical reaction conditions for enhanced production yields
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摘要（原文节选）

Finding optimal reaction conditions is crucial for chemical synthesis in the pharmaceutical and chemical industries. However, due to the vast chemical space, conducting experiments for all the possible combinations is impractical. Thus, quantitative structure – activity relationship (QSAR) models have been widely used to predict product yields, but evaluating all combinations is still computationally intensive. In this work, we demonstrate the use of Digital Annealer Unit (DAU) can tackle these large-scale optimizati ⋯

数据表 · Table 3（6 行，含数值行 5 行）

Results for the models trained on three subsets selected from s1, s3, s7, and s9 in HTE datasets, evaluated on the held-out test set of the C-N cross-coupling dataset with one-hot encoding via ML-based and DAU-based model.

			s1					s3
	ML	FFN	SVM	RF	DAU	ML	FFN	SVM
Top-5 accuracy score	0	0	2	1	0	1	1	0
Top-10 accuracy score	0	0	3	1	0	3	4	3
Top-15 accuracy score	0	0	5	2	0	5	6	5
Top-20 accuracy score	0	0	8	5	0	8	7	7
Pearson correlation coefficient	0.09	0.10	0.50	0.56	-0.03	0.83	0.84	0.69

脚注（逐字）：Top-k score and Pearson correlation coefficient are used as metrics. The top-k scores in s3, s7 and s9 show the possibility that a QUBO-based model can be used for optimal yield prediction, meaning that the model could recommend reaction conditions corresponding to high yields in real experiments

正文对应句（逐字）：In this work, DAU is used to optimize reaction conditions to maximize production yields since it is difficult to solve with existing general-purpose computers. In order to use DAU, a problem needs to be formulated as a QUBO form. The mathematical form is expressed as:1\documentclass[12pt]{minimal} \

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